

PAPER

nf-core/viralmetagenome: A Novel Pipeline for Untargeted Viral Genome Reconstruction

Joon Klaps^{1,*}, Philippe Lemey¹, Magda Bletsas¹, nf-core community² and Liana Eleni Kafetzopoulou^{1,3}

¹Rega Institute for Medical Research, Department of Microbiology, Immunology and Transplantation, KU Leuven, Herestraat 49, 3000, Leuven, Belgium, ²A full list of contributors can be found at <https://nf-co.re/community>. and ³ Leiden University Center of Infectious Diseases (LUCID), Leiden University Medical Center, Albinusdreef 2, 2333 ZA, Leiden, the Netherlands

*Corresponding author. joon.klaps@kuleuven.be

FOR PUBLISHER ONLY Received on Date Month Year; revised on Date Month Year; accepted on Date Month Year

Abstract

Motivation: Reconstructing eukaryotic viral genomes from metagenomic data is challenging due to their extensive diversity and potential genome segmentation. Current approaches often rely on labor-intensive manual curation for reference selection and scaffolding, limiting scalability for large studies or rapid outbreak response. We address the critical need for an automated, scalable pipeline for efficient viral metagenomic analysis without manual intervention.

Results: We present nf-core/viralmetagenome, a comprehensive Nextflow pipeline for the untargeted reconstruction and variant analysis of eukaryotic DNA and RNA viruses from short-read metagenomic or hybridisation capture enriched samples. The pipeline automates the entire process from read preprocessing to consensus generation, integrating multiple *de novo* assemblers, automated reference selection, and iterative consensus refinement. It features robust quality control, extensive documentation, and seamless portability via Docker and Singularity. We validated the pipeline on diverse simulated and real datasets, demonstrating its ability to recover high-quality genomes from complex metagenomic samples and resolve co-infections, making it a powerful tool for viral surveillance.

Availability: nf-core/viralmetagenome is freely available at <https://github.com/nf-core/viralmetagenome> with comprehensive documentation at <https://nf-co.re/viralmetagenome>. Archival code repository snapshots are published at zenodo with doi: <https://doi.org/10.5281/zenodo.17524074>.

Contact: joon.klaps@kuleuven.be

Supplementary information: Supplementary data are available at <https://github.com/Joon-Klaps/nf-core-viralmetagenome-manuscript> online.

Key words: viralmetagenome, bioinformatic pipeline, nextflow, viral metagenomics, viral assembly, viral variant analysis

Introduction

Reconstructing viral genomes from metagenomic sequencing data presents considerable computational challenges, particularly for viruses exhibiting extensive genetic diversity (Baaijens et al. 2017, Meleshko et al. 2021). This challenge is amplified in segmented viruses such as influenza, rotavirus, and bunyaviruses, where individual segments can evolve under distinct selective pressures and can undergo reassortment. While some pipelines target specific viruses and their subtypes (Shepard et al. 2016), accurate genome reconstruction from samples with unknown viral strains typically requires time-consuming manual curation of contigs and reference matching (de Vries et al. 2021). This makes large-scale metagenome studies or rapid outbreak responses impractical.

To address these limitations, we developed nf-core/viralmetagenome, a comprehensive pipeline specifically designed for untargeted viral genome reconstruction using short read sequences. The pipeline is developed using Nextflow (Di Tommaso et al. 2017) within the nf-core framework (Langer et al. 2025), ensuring reproducibility through containerization with Docker (Merkel 2014) and Singularity (Kurtzer et al. 2017), as well as portability across computational platforms such as local desktops, high-performance clusters and cloud environments.

Pipeline Description

nf-core/viralmetagenome implements a workflow for *de novo* assembly, reference matching, and iterative consensus refinement to reconstruct viral genomes without prior

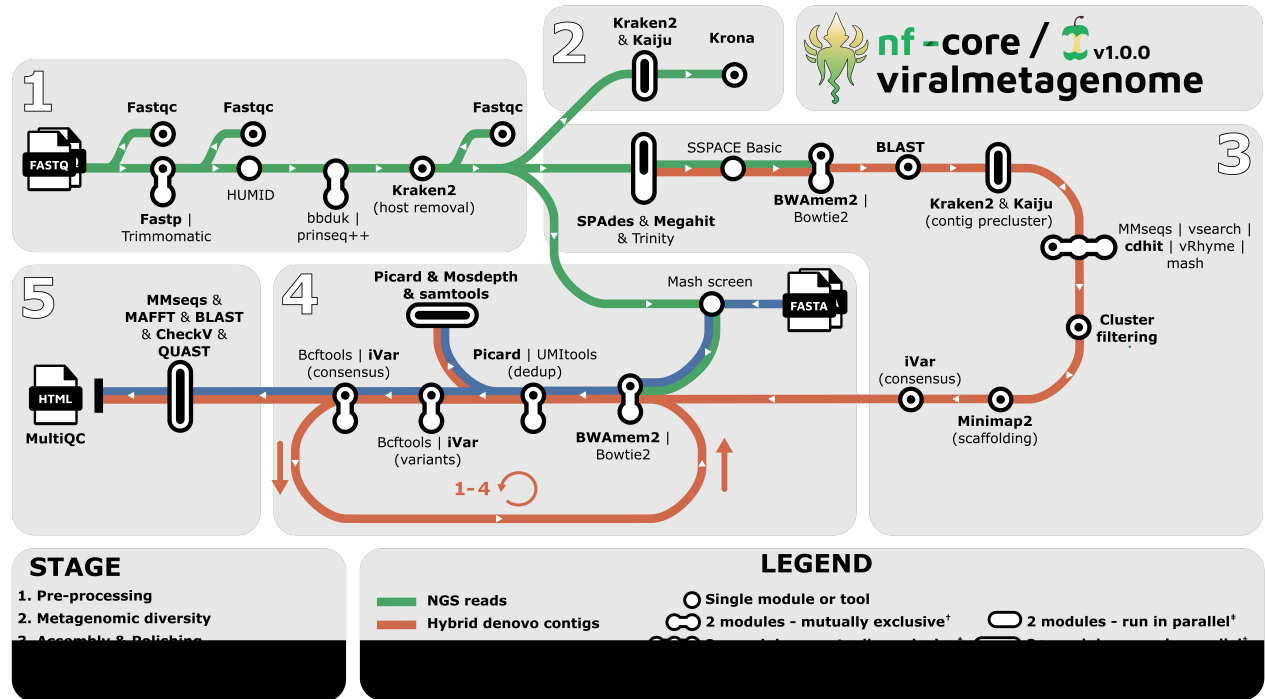


Fig. 1. Overview of the nf-core/viralmetagenome pipeline for untargeted viral genome reconstruction. Default options are highlighted in bold. nf-core/viralmetagenome processes short-read data through pre-processing, metagenomic diversity assessment, *de novo* assembly with multiple assemblers, scaffolding with automated reference identification and taxonomy-guided contig clustering, and iterative consensus refinement. Quality-control metrics, assembly statistics, and coverage data are summarised in MultiQC reports and overview tables for downstream analysis.

target knowledge. The pipeline comprises five stages: read preprocessing, metagenomic diversity assessment, contig assembly and scaffolding, iterative consensus refinement with variant analysis, and quality control (Figure 1). Unless otherwise specified, the pipeline offers multiple options to accommodate established user workflows and preferences. Detailed tool descriptions are provided in Supplementary Table S1.

Read preprocessing

Input reads are provided via sample sheets containing sample names and paths to short-read FASTQ files. These are preprocessed using FastQC and trimmed for adapters using fastp (Chen et al. 2018) (default) or Trimmomatic (Bolger et al. 2014). Fastp offers automated adapter detection and faster processing (Chen et al. 2018). UMI deduplication is implemented using HUMID (GitHub repository: <https://github.com/jfjlaros/HUMID>) for raw reads, or UMI-tools after mapping (Smith et al. 2017). Multiple sequencing runs can be merged after trimming by specifying merge group identifiers in the sample sheet. Optional filtering with bbdduk (Bushnell 2014) or PRINSEQ++ (Cantu et al. 2019) removes low-complexity sequences, while host reads are removed with Kraken2 (Wood et al. 2019).

Metagenomic diversity assessment

Preprocessed reads are taxonomically classified with Kaiju (Menzel et al. 2016) and Kraken2 (Wood et al. 2019) to maximise sensitivity across diverse viral families. Results from both classifiers are visualised using Krona (Ondov et al. 2011).

De novo assembly and clustering

The assembly workflow employs a multi-assembler approach where contigs generated by all tools are pooled and processed jointly. *De novo* assembly uses SPAdes (Meleshko et al. 2021) (RNAviral mode), MEGAHIT (Li et al. 2016), and Trinity (Grabherr et al. 2011), capitalizing on their distinct algorithmic strengths to maximise genome recovery across diverse viral families and read depths. Subsequently, a BLASTn (Altschul et al. 1990) search against the Reference Viral Database (RVDB) (Chin et al. 2025) identifies candidate reference genomes for each contig. All contigs and their identified reference genomes are subjected to clustering. This strategy enables contig scaffolding against the cluster centroid whether a database reference genome or a *de novo* contig, thereby combining outputs from multiple assemblers into unified scaffolds.

Clustering is performed in two sequential stages. First, taxonomic pre-clustering groups contigs based on taxonomic classification using Kraken2 (Wood et al. 2019) and Kaiju (Menzel et al. 2016), with optional filtering to focus on specific taxonomic clades for more targeted analyses. Second, nucleotide similarity clustering within taxonomic groups is performed using CD-HIT-EST (Li and Godzik 2006) (default), VSEARCH (Rognes et al. 2016), MMseqs2 (Steinegger and Söding 2017), vRhyme (Kieft et al. 2022), or Mash (Ondov et al. 2019). All tools are valid options, though performance may vary by dataset (Zielezinski et al. 2025, Steinegger and Söding 2017).

Optionally, following *de novo* assembly and extension, reads can be mapped to all contigs using BWA-MEM2 (Vasimuddin et al. 2019) (default), or Bowtie2 (Langmead et al. 2019).

Clusters are filtered based on the total percentage of reads mapping to all contigs within a cluster, removing assembly artefacts. For the final scaffolding step, all cluster members are mapped to the cluster representative or centroid using Minimap2 (Li 2018), followed by consensus calling with iVar (Grubaugh et al. 2019) to generate reference-assisted scaffolds.

Consensus generation and variant calling

The consensus module supports reference-guided analysis and scaffold refinement using user-defined reference genomes. Users can provide one or multiple references; when multiple are provided, Mash (Ondov et al. 2019) selects the most similar one. The selected reference then undergoes a single refinement round to generate the consensus.

In contrast, scaffold refinement employs an iterative approach, performing up to 4 cycles (default 2) to polish the consensus sequence. Both processes map reads using BWA-MEM2 (Vasimuddin et al. 2019) (default), or Bowtie2 (Langmead et al. 2019), followed by variant calling with BCFtools (Danecek et al. 2021) or iVar (Grubaugh et al. 2019) (default). Benchmarking (Bassano et al. 2022) indicated that BCFtools outperforms iVar in precision and recall, while iVar identified more low-frequency variants. Users can prioritise sensitivity or specificity when selecting the variant caller. Final variant calling is performed on the polished consensus to characterise single nucleotide variants.

Consensus Quality Control

Quality control employs CheckV (Nayfach et al. 2021) to assess completeness, BLASTn (Altschul et al. 1990) for reference similarity, and MMseqs2 (Steinegger and Söding 2017) against the annotated database Virosaurus (Gleizes et al. 2020). These analyses enable species identification, genomic segment classification, host prediction, and retrieval of additional metadata embedded within the reference databases.

The consensus refinement is evaluated through sequence alignment with MAFFT (Katoh et al. 2002), which compares final consensus genomes against *de novo* contigs, intermediate consensus sequences from iterative cycles, and the scaffolding reference. All tool metrics are compiled into an interactive MultiQC report (Ewels et al. 2016). Additionally, key metrics are extracted from the MultiQC report and compiled into overview tables for downstream analysis.

Applications

We validated the pipeline’s performance using a diverse set of real metagenomic samples containing human and plant pathogens. nf-core/viralmetagene successfully reconstructed high-quality or near-complete consensus genomes across various viral families, including segmented viruses (Lassa virus, Orthornairovirus, Tomato spotted wilt tospovirus) and non-segmented viruses (SARS-CoV-2, West Nile virus, Potato virus Y, Youcai mosaic virus, and Monkeypox virus; Supplementary methods S1). The pipeline proved efficient, processing 28 samples in 412 CPU hours with a maximum memory footprint of 79GB on a high-performance cluster. To facilitate local deployment, we implemented a `-profile local` option, which utilizes a smaller viral RefSeq database instead of the full RVDB database for Kaiju, reducing the pipeline’s peak RAM usage to approximately 18GB.

To evaluate the pipeline’s robustness in complex scenarios, we conducted two targeted experiments. First, we assessed

its ability to resolve co-infections using a simulated mixture of viral genomes (Supplementary Methods S2). The pipeline successfully distinguished and reconstructed distinct co-infecting strains, provided the genetic divergence was sufficient ($\leq 88.7\%$ average nucleotide identity (ANI); Figure 2).

To assess performance, we compared all generated consensus sequences with the two reference genomes used to generate the synthetic read mixture (the constant reference genome MN090277.1 and the representative genome for each bin). We performed global alignments using MAFFT (Katoh et al. 2002) and determined their alignment statistics (number of exact matches, approximate matches, number of mismatches, and alignment length). For each generated consensus, we determined the closest matching reference (i.e. the constant reference genome MN090277.1 or the representative reference genome) based on the highest number of exact matches in the global alignment. We identified three distinct performance zones based on empirical ANI thresholds (Figure 2):

- **Unresolved Zone ($\geq 88.7\%$):** When the genetic similarity between the constant reference sequence MN090277.1 and the representative reference genome in the mixture is relatively high, the pipeline is unable to distinguish which reference genome the consensus sequence corresponds to. This leads to a single mixed consensus sequence for each interval bin in this high genetic similarity range.
- **Ambiguous Zone ($85.8\% - 88.7\%$):** In this intermediate zone, the pipeline successfully resolves the simulated mixture; however, the sequence similarity still permits reads to cross-map between references. This causes the reconstructed consensus sequences to accumulate mismatches in shared regions, resulting in lower fidelity to the original reference genomes despite initial separation by the assemblers.
- **High-Fidelity Resolution ($< 85.8\%$):** Below this similarity threshold, the reference genomes used to create the simulated mixture are sufficiently divergent to minimize cross-mapping interference, allowing the pipeline to accurately reconstruct distinct, high-quality consensus genomes.

The second part of our benchmarking examined how scaffolding-reference similarity affects consensus genome reconstruction. We selected sequencing data from seven Lassa virus-positive human clinical samples with diverse viral loads (Supplementary Methods S3; Supplementary Fig. S1). In these samples, we observed significantly higher sequencing depth for the S segment than for the L segment (Figure 3). To establish a benchmark, we generated a baseline consensus genome for each sample using automated reference selection from the full unclustered-RVDB (U-RVDB) reference pool. We then systematically challenged the scaffolding process by providing reference genomes (19-30 per sample; Table 1) spanning a range of similarity to the baseline consensus (64%–100% ANI; Supplementary Fig. S2). For each supplied reference, we generated a new consensus genome using it as the scaffolding backbone when clustered with the *de novo* contigs.

Comparing these sequences against their baseline consensus, we found that reference scaffolding improved reconstruction (Figure 4), particularly in low read depth scenarios where *de novo* assembly struggled. Here, we defined consensus accuracy as the Average Nucleotide Identity between each test consensus and its baseline (Supplementary Methods S3). For the high-coverage S segment (Figure 3), *de novo* assembly



Fig. 2. Mixture resolution dynamics across a genetic diversity gradient. This scatter plot characterises nf-core/viralmetagenome’s ability to resolve co-infections of varying genetic similarity. The y-axis (Representative vs Constant Similarity) represents the ANI (Eq. 1) between the representative genome and the constant genome MN090277.1. The x-axis shows the ANI between each reconstructed consensus genome and the genome represented in the corresponding panel: the representative genome in the left panel and the constant genome MN090277.1 in the right panel. Blue points in the left panel are consensus genomes whose best match is the representative genome; green points in the right panel are consensus genomes whose best match is the constant genome. Triangles represent a failure in resolving the mixture as only a single genome was reconstructed, circles represent a successful identification of both the constant genome and the representative genome. A perfect mixture resolution yields a blue point near 100% ANI in the representative genome panel and a green point near 100% ANI in the constant genome panel. Three empirically determined zones are highlighted: Unresolved ($>88.7\%$ Representative vs Constant Similarity), Ambiguous ($85.8\text{--}88.7\%$), and High-Fidelity Resolution ($<85.8\%$).

Table 1. Overview of the number of selected references used for each sample to determine the influence of scaffolding reference (Figure 4; Supplementary Fig. S2). References were systematically selected to span a gradient of similarity to each sample’s baseline consensus genome, with one reference selected per 2% ANI interval (e.g. 64–66%, 66–68%, ..., 98–100%).

Sample ID	Total References	L Segment	S Segment
LVE00045	22	13	9
LVE00050	19	7	12
LVE00063	20	11	9
LVE00127	30	16	14
LVE00140	23	13	10
LVE00167	21	11	10
LVE00279	22	12	10

was sufficient to reconstruct the full genome, rendering the choice of scaffolding reference largely redundant (Figure 4A). In contrast, the low-coverage L segment proved more challenging: *de novo* assembly produced shorter, incomplete fragments, and the pipeline needed a closely related database reference genome to bridge gaps and recover a complete segment consensus sequence.

These findings emphasise the need to keep databases like RVDB (Chin et al. 2025) up-to-date. Since nf-core/viralmetagenome is primarily designed for eukaryotic

viruses, users interested in bacteriophage analysis are encouraged to explore pipelines designed to target phages such as VIRify (Rangel-Pineros et al. 2022), VIBRANT (Kieft et al. 2020), VirSorter2 (Guo et al. 2021).

Conclusion

nf-core/viralmetagenome addresses a critical need in viral genomics by providing an automated, scalable solution for untargeted viral genome reconstruction. The pipeline successfully automates the traditionally time-consuming and manual process of viral genome reconstruction from short-read metagenomic data through its integrated workflow of *de novo* contig assembly, automated reference selection, clustering algorithms, and iterative refinement strategies.

Our validation results indicate that the pipeline is suitable for a wide range of eukaryotic viral families, it removes the burden of extensive manual curation, and enables consistent, high-quality genome reconstruction with reproducible results across varied computational settings. As viral surveillance and outbreak response increasingly rely on metagenomic sequencing, automated pipelines like nf-core/viralmetagenome will be essential for the timely identification of pathogen strains. The pipeline represents a significant step forward in making viral genome reconstruction accessible to researchers without

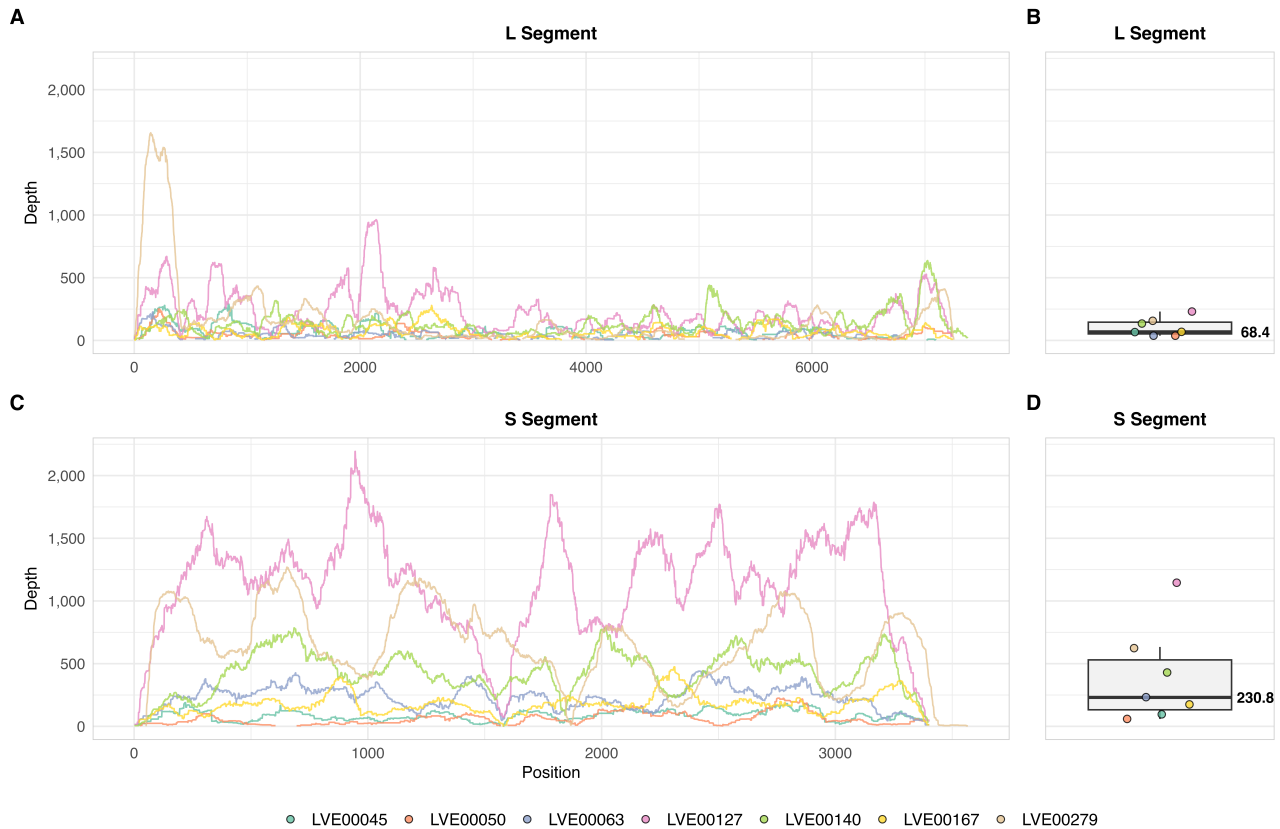


Fig. 3. Sequencing coverage across Lassa virus baseline consensus genomes. Consensus genomes used to determine the coverages are the baseline consensus genomes (generated with U-RVDB as a reference library supplied with `--reference_pool`). A,C display per-base coverage depth ($\log_{10}$ scale) along the genome. B,D display mean coverage per consensus. Overall, the S segment has a high read depth (median $\sim 231X$), whereas the L segment shows lower coverage (median $\sim 68X$) along with multiple areas of zero coverage, presenting a challenge for de novo assembly.

requiring extensive bioinformatics expertise, facilitating broader adoption of metagenomic approaches in viral research and public health applications.

Acknowledgments

J.K, P.L. and L.E.K acknowledge support from the Research Foundation - Flanders (Fonds voor Wetenschappelijk Onderzoek – Vlaanderen, G005323N and G051322N, 1SH2V24N, 12X9222N). Assistance from AI language models (e.g., ChatGPT) was used to enhance the clarity of the manuscript and to support code development.

Author Contributions

J.K. designed and implemented the pipeline, performed validation analyses, and wrote the manuscript. P.L. and L.E.K. supervised the project, and M.B. provided critical feedback. The nf-core community contributed to pipeline maintenance. All authors reviewed and approved the final manuscript.

Conflict of Interest

The authors declare no competing interests.

References

- S F Altschul, et al. Basic local alignment search tool. *J. Mol. Biol.*, 215(3):403–410, 1990. doi: 10.1016/S0022-2836(05)80360-2.
- Jasmijn A Baaijens, Amal Zine El Aabidine, Eric Rivals, and Alexander Schönhuth. De novo assembly of viral quasiespecies using overlap graphs. *Genome Res.*, 27(5):835–848, 2017. doi: 10.1101/gr.215038.116.
- Irene Bassano, et al. Evaluation of variant calling algorithms for wastewater-based epidemiology using mixed populations of SARS-CoV-2 variants in synthetic and wastewater samples. *bioRxiv*, 2022. doi: 10.1101/2022.06.06.22275866.
- Anthony M Bolger, Marc Lohse, and Bjoern Usadel. Trimmomatic: a flexible trimmer for illumina sequence data. *Bioinformatics*, 30(15):2114–2120, 2014. doi: 10.1093/bioinformatics/btu170.
- Brian Bushnell. BBMap: A Fast, Accurate, Splice-Aware Aligner, 2014.
- Vito Adrian Cantu, Jeffrey Sadural, and Robert Edwards. PRINSEQ++, a multi-threaded tool for fast and efficient quality control and preprocessing of sequencing datasets. *PeerJ*, 2019. doi: 10.7287/peerj.preprints.27553v1.
- Shifu Chen, Yanqing Zhou, Yaru Chen, and Jia Gu. fastp: an ultra-fast all-in-one FASTQ preprocessor. *Bioinformatics*, 34(17):i884–i890, 2018. doi: 10.1093/bioinformatics/bty560.

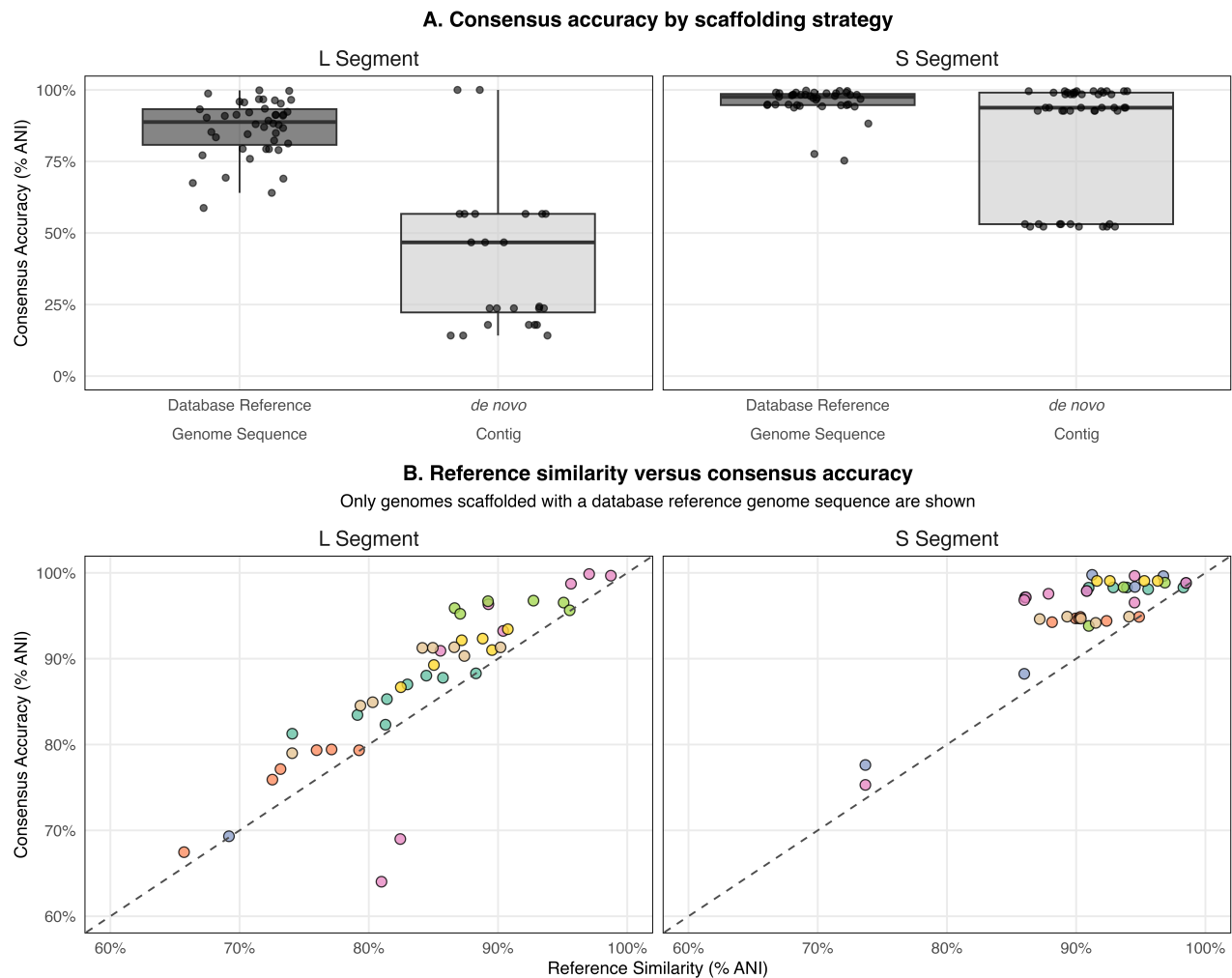


Fig. 4. Impact of scaffolding reference similarity on consensus reconstruction accuracy. Consensus accuracy is defined as the ANI (Eq. 1 in Supplementary Methods) between the test consensus and the baseline consensus (generated using the pipeline's automated reference selection from the full unclustered-RVDB). Test consensus genomes were scaffolded using either a supplied database reference genome or a *de novo* contig. **(A)** Boxplots comparing Consensus Accuracy (% ANI) for the L and S segments. The plots contrast the performance of using a database reference genome as the scaffolding backbone (dark grey) against relying solely on the *de novo* contig (light grey). For the low-coverage L segment, using the database reference genome significantly improves accuracy, whereas for the high-coverage S segment, the *de novo* route performs comparably to the database reference genome scaffolding (sequencing depths detailed in Figure 3). **(B)** Scatter plots illustrate the relationship between reference similarity and consensus accuracy (% ANI) for the L and S segments. Here, reference similarity is defined as the ANI between the supplied database reference genome sequence and the baseline consensus genome. The dashed line represents the line of identity ($y = x$), and points are coloured by sample source (Supplementary Fig. S1). Only consensus genomes that used a database reference genome sequence as the scaffold reference are shown in panel B.

Pei-Ju Chin, et al. Refinement of the reference viral database (RVDB) for improving bioinformatics analysis of virus detection by high-throughput sequencing (HTS). *mSphere*, 10(7):e0028625, 2025. doi: 10.1128/msphere.00286-25.

Petr Danecek, et al. Twelve years of SAMtools and BCFtools. *Gigascience*, 10(2), 2021. doi: 10.1093/gigascience/giab008.

Jutte J C de Vries, et al. Benchmark of thirteen bioinformatic pipelines for metagenomic virus diagnostics using datasets from clinical samples. *J. Clin. Virol.*, 141:104908, 2021. doi: 10.1016/j.jcv.2021.104908.

Paolo Di Tommaso, et al. Nextflow enables reproducible computational workflows. *Nat. Biotechnol.*, 35(4):316–319, 2017. doi: 10.1038/nbt.3820.

Philip Ewels, Måns Magnusson, Sverker Lundin, and Max Källér. MultiQC: summarize analysis results for multiple

tools and samples in a single report. *Bioinformatics*, 32(19):3047–3048, 2016. doi: 10.1093/bioinformatics/btw354.

Anne Gleizes, et al. Virosaurus a reference to explore and capture virus genetic diversity. *Viruses*, 12(11), 2020. doi: 10.3390/v12111248.

Manfred G Grabherr, et al. Full-length transcriptome assembly from RNA-seq data without a reference genome. *Nat. Biotechnol.*, 29(7):644–652, 2011. doi: 10.1038/nbt.1883.

Nathan D Grubaugh, et al. An amplicon-based sequencing framework for accurately measuring intrahost virus diversity using PrimalSeq and iVar. *Genome Biol.*, 20(1):8, 2019. doi: 10.1186/s13059-018-1618-7.

Jiarong Guo, et al. VirSorter2: a multi-classifier, expert-guided approach to detect diverse DNA and RNA viruses. *Microbiome*, 9(1):37, 2021. doi: 10.1186/s40168-020-00990-y.

- Kazutaka Katoh, Kazuharu Misawa, Kei-Ichi Kuma, and Takashi Miyata. MAFFT: a novel method for rapid multiple sequence alignment based on fast fourier transform. *Nucleic Acids Res.*, 30(14):3059–3066, 2002. doi: 10.1093/nar/gkf436.
- Kristopher Kieft, et al. vRhyme enables binning of viral genomes from metagenomes. *Nucleic Acids Res.*, 50(14): e83, 2022. doi: 10.1093/nar/gkac341.
- Kristopher Kieft, Zhichao Zhou, and Karthik Anantharaman. VIBRANT: automated recovery, annotation and curation of microbial viruses, and evaluation of viral community function from genomic sequences. *Microbiome*, 8(1):90, 2020. doi: 10.1186/s40168-020-00867-0.
- Gregory M Kurtzer, Vanessa Sochat, and Michael W Bauer. Singularity: Scientific containers for mobility of compute. *PLoS One*, 12(5):e0177459, 2017. doi: 10.1371/journal.pone.0177459.
- Björn E Langer, et al. Empowering bioinformatics communities with nextflow and nf-core. *Genome Biol.*, 26(1):228, 2025. doi: 10.1186/s13059-025-03673-9.
- Ben Langmead, Christopher Wilks, Valentin Antonescu, and Rone Charles. Scaling read aligners to hundreds of threads on general-purpose processors. *Bioinformatics*, 35(3):421–432, 2019. doi: 10.1093/bioinformatics/bty648.
- Dinghua Li, et al. MEGAHIT v1.0: A fast and scalable metagenome assembler driven by advanced methodologies and community practices. *Methods*, 102:3–11, 2016. doi: 10.1016/j.jymeth.2016.02.020.
- Heng Li. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics*, 34(18):3094–3100, 2018. doi: 10.1093/bioinformatics/bty191.
- Weizhong Li and Adam Godzik. Cd-hit: a fast program for clustering and comparing large sets of protein or nucleotide sequences. *Bioinformatics*, 22(13):1658–1659, 2006. doi: 10.1093/bioinformatics/btl158.
- Dmitry Meleshko, Iman Hajirasouliha, and Anton Korobeynikov. coronaSPAdes: from biosynthetic gene clusters to RNA viral assemblies. *Bioinformatics*, 2021. doi: 10.1093/bioinformatics/btab597.
- Peter Menzel, Kim Lee Ng, and Anders Krogh. Fast and sensitive taxonomic classification for metagenomics with kaiju. *Nat. Commun.*, 7:11257, 2016. doi: 10.1038/ncomms11257.
- D Merkel. Docker: lightweight linux containers for consistent development and deployment. *Linux Journal*, 2014(239):2, 2014. doi: 10.5555/2600239.2600241.
- Stephen Nayfach, et al. CheckV assesses the quality and completeness of metagenome-assembled viral genomes. *Nat. Biotechnol.*, 39(5):578–585, 2021. doi: 10.1038/s41587-020-00774-7.
- Brian D Ondov, Nicholas H Bergman, and Adam M Phillippy. Interactive metagenomic visualization in a web browser. *BMC Bioinformatics*, 12(1):385, 2011. doi: 10.1186/1471-2105-12-385.
- Brian D Ondov, et al. Mash screen: high-throughput sequence containment estimation for genome discovery. *Genome Biol.*, 20(1):232, 2019. doi: 10.1186/s13059-019-1841-x.
- Guillermo Rangel-Pineros, et al. VIRify: an integrated detection, annotation and taxonomic classification pipeline using virus-specific protein profile hidden markov models. *bioRxiv*, page 2022.08.22.504484, 2022. doi: 10.1101/2022.08.22.504484.
- Torbjørn Rognes, et al. VSEARCH: a versatile open source tool for metagenomics. *PeerJ*, 4:e2584, 2016. doi: 10.7717/peerj.2584.
- Samuel S Shepard, et al. Viral deep sequencing needs an adaptive approach: IRMA, the iterative refinement meta-assembler. *BMC Genomics*, 17(1):708, 2016. doi: 10.1186/s12864-016-3030-6.
- Tom Smith, Andreas Heger, and Ian Sudbery. UMI-tools: modeling sequencing errors in unique molecular identifiers to improve quantification accuracy. *Genome Res.*, 27(3): 491–499, 2017. doi: 10.1101/gr.209601.116.
- Martin Steinegger and Johannes Söding. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nat. Biotechnol.*, 35(11):1026–1028, 2017. doi: 10.1038/nbt.3988.
- Md Vasimuddin, Sanchit Misra, Heng Li, and Srinivas Aluru. Efficient architecture-aware acceleration of BWA-MEM for multicore systems. In *2019 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 314–324. IEEE, 2019. doi: 10.1109/IPDPS.2019.00041.
- Derrick E Wood, Jennifer Lu, and Ben Langmead. Improved metagenomic analysis with kraken 2. *Genome Biol.*, 20(1): 257, 2019. doi: 10.1186/s13059-019-1891-0.
- Andrzej Zieleszinski, et al. Ultrafast and accurate sequence alignment and clustering of viral genomes. *Nat. Methods*, 22(6):1191–1194, 2025. doi: 10.1038/s41592-025-02701-7.